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Wheeled carrying apparatus

Abstract

A wheeled carrying apparatus includes a standing frame, a handle frame, a wheel mount, a locking assembly and a release mechanism. The handle frame is rotatable between a first and a second position. The wheel mount carries a wheel and is rotatable relative to the standing frame for changing an orientation of the wheel axis. The locking assembly is configured to rotationally lock the wheel mount with respect to the standing frame, and configured to unlock the wheel mount so that the wheel mount is rotatable relative to the standing frame for changing an orientation of the wheel axis. The release mechanism includes a first portion provided on the standing frame and coupled to the locking mechanism, and a second portion carried with the handle frame, the second portion being movable along with the handle frame toward and away from the first portion during rotation of the handle frame.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/945,323, filed on Sep. 15, 2022, which is a continuation application of U.S. patent application Ser. No. 17/017,401, filed on Sep. 10, 2020, now U.S. Pat. No. 11,479,285, issued on Oct. 25, 2022, which claims priority to Chinese Patent Application No. 201910858955.3, filed on Sep. 11, 2019, and to Chinese Patent Application No. 202010370191.6, filed on Apr. 30, 2020, which are incorporated by reference as if fully set forth.

FIELD OF THE INVENTION

(1) The present invention relates to wheeled carrying apparatuses.

BACKGROUND OF THE INVENTION

(2) Caregivers usually rely on a stroller apparatus to transport babies and children. Most stroller apparatuses typically have front wheels that can freely rotate horizontally relative to the stroller frame for changing the orientation of the wheel axes, and rear wheels that cannot rotate horizontally and have a fixed wheel axis oriented transversally. As a result, it may not be easy for the stroller apparatus to turn around in an environment of limited space.

(3) Therefore, there is a need for an improved design that can offer better maneuverability and address at least the foregoing issues.

SUMMARY OF THE INVENTION

(4) The present application describes a wheeled carrying apparatus having wheel assemblies that can be controllably locked and unlocked in horizontal orientation and can offer better maneuverability.

(5) According to one embodiment, the wheeled carrying apparatus includes a standing frame, a handle frame, a first wheel mount, a locking assembly and a release mechanism. The standing frame and the handle frame are pivotally coupled to each other, the handle frame being rotatable

relative to the standing frame between a first position and a second position. The first wheel mount carries a first wheel and is pivotally connected with the standing frame, the first wheel being rotatable relative to the first wheel mount about a first wheel axis. The locking assembly is connected with the standing frame, the locking assembly being configured to rotationally lock the first wheel mount with respect to the standing frame, and configured to unlock the first wheel mount so that the first wheel mount is rotatable relative to the standing frame for changing an orientation of the first wheel axis. The release mechanism includes a first portion provided on the standing frame and coupled to the locking assembly, and a second portion carried with the handle frame, the second portion being movable along with the handle frame during rotation of the handle frame between the first position and the second position. The second portion is operable to actuate the first portion and cause the locking assembly to unlock the locking assembly when the handle frame is in the first position, and the second portion is decoupled from the first portion so that the second portion is inoperable to cause the locking assembly to unlock the first wheel mount when the handle frame is in the second position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view illustrating an embodiment of a wheeled carrying apparatus;
- (2) FIG. 2 is a cross-sectional view illustrating one wheel assembly provided in the wheeled carrying apparatus;
- (3) FIG. 3 is an enlarged perspective view illustrating another wheel assembly provided in the wheeled carrying apparatus;
- (4) FIG. 4 is a side view illustrating some construction details of an actuating assembly provided in the wheeled carrying apparatus;
- (5) FIG. 5 is a perspective view illustrating some construction details of the actuating assembly and a release mechanism provided in the wheeled carrying apparatus;
- (6) FIG. 6 is a side view illustrating some construction details of the release mechanism provided in the wheeled carrying apparatus;
- (7) FIG. 7 is a perspective view illustrating a portion of the actuating assembly and release mechanism connected with a carrier bar in the frame structure of the wheeled carrying apparatus;
- (8) FIG. 8 is an enlarged view illustrating another portion of the release mechanism provided on a leg frame of the wheeled carrying apparatus;
- (9) FIG. 9 is an enlarged view illustrating construction details of an operating part provided on a handle frame of the wheeled carrying apparatus;
- (10) FIG. 10 is a side view illustrating a variant construction of the release mechanism;
- (11) FIG. 11 is a perspective view illustrating some construction details of the release mechanism shown in FIG. 10; and
- (12) FIG. 12 is a perspective view illustrating the wheeled carrying apparatus with the handle frame inclined in a second position different from that shown in FIG. 1.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) The present application provides a wheeled carrying apparatus that includes a standing frame, and a plurality of wheel assemblies provided at a bottom of the standing frame. The wheel assemblies can be controllably locked and unlocked in horizontal orientation with respect to a standing frame, wherein an unlocked wheel assembly can rotate relative to the standing frame for changing the horizontal orientation of a wheel axis. According to the needs, some of the wheel assemblies are locked while others are unlocked, or all of the wheel assemblies can be unlocked at the same time as desired. Accordingly, the wheeled carrying apparatuses can be more flexible in use and have better maneuverability. Embodiments of the wheeled carrying apparatus described

herein include child stroller apparatuses. However, it will be appreciated that wheeled carrying apparatuses that may incorporate the structures and features described herein can include, without limitation, child stroller apparatuses, trolleys, shopping carts, and the like. Accordingly, it is understood that the structures and features described herein may be generally applicable to any types of products having wheels, and the wheeled carrying apparatuses as described and claimed herein are not limited to the examples illustrated hereinafter.

(14) FIG. 1 is a perspective view illustrating an embodiment of a wheeled carrying apparatus **100**, which is exemplarily a child stroller apparatus. Referring to FIG. 1, the wheeled carrying apparatus **100** can include a frame structure **102** comprised of a standing frame **104** and a handle frame **106** coupled to each other, and a plurality of wheel assemblies **108** and **110** provided at a bottom of the standing frame **104**.

(15) According to an example of construction, the standing frame **104** can include two leg frames **112** and **114**, two side linking bars **116**, two side support bars **118** and two carrier bars **120**. The leg frame **112** can be exemplarily a rear leg frame, and the leg frame **114** can be exemplarily a front leg frame. Each of the leg frames **112** and **114** can respectively include two side segments disposed symmetrically at a left and a right side of the standing frame **104**, e.g., two side segments **112A** for the leg frame **112** and two side segments **114A** for the leg frame **114**, and a transversal segment fixedly connected with the two side segments, e.g., a transversal segment **112B** fixedly connected with the two side segments **112A** for the leg frame **112**, and a transversal segment **114B** fixedly connected with the two side segments **114A** for the leg frame **114**. At each of the left and right sides, the side segment **112A** of the leg frame **112** and the side segment **114A** of the leg frame **114** can be respectively coupled pivotally to one side linking bar **116**, and one side support bar **118** can extend between side segments **112A** and **114A** and can be pivotally coupled to the side segment **114A**, whereby the leg frames **112** and **114**, the side linking bars **116** and the side support bars **118** can rotate relative to one another during folding and unfolding of the wheeled carrying apparatus **100**.

(16) The handle frame **106** can be respectively coupled pivotally to the standing frame **104** at the left and right sides via two bar linkages **122**, wherein each bar linkage **122** can include a bar **122A**. More specifically, the handle frame **106** can include two side portions **106A** disposed symmetrically at the left and right sides, and a grip portion **106B** connected with the two side portions **106A**. At each of the left and right sides, the carrier bar **120** can have an upper portion pivotally connected with the side linking bar **116**, and a lower portion respectively connected pivotally with the side support bar **118**, the bar **122A** of the bar linkage **122** and the side portion **106A** of the handle frame **106** about a common pivot axis R. The bar **122A** of the bar linkage **122** can be respectively connected pivotally with the side segment **112A** of the leg frame **112**, and with the side portion **106A** of the handle frame **106** about the pivot axis R. With this construction, the handle frame **106** can rotate about the pivot axis R relative to the standing frame **104** and the bar linkage **122** for adjustment between the two positions of different inclinations so that the wheeled carrying apparatus **100** can be pushed in different directions with a child facing forward or rearward. For example, the handle frame **106** can be inclined toward the side of the leg frame **112** in a first position as shown in FIG. 1, and inclined toward the side of the leg frame **114** in a second position as shown in FIG. 12.

(17) Referring to FIG. 1, each of the wheel assemblies **108** and **110** can include a wheel mount that is pivotally connected with the standing frame **104** and carries at least a wheel rotatable relative to the wheel mount about a wheel axis extending generally horizontally, whereby the wheel mount is rotatable relative to the standing frame **104** for changing an orientation of the wheel axis. For example, two wheel assemblies **108** can be provided at the lower ends of the side segments **112A** of the leg frame **112**, and two wheel assemblies **110** can be respectively provided at the lower ends of the two side segments **114A** of the leg frame **114**. Each wheel assembly **108** can include a wheel mount **124** that is pivotally connected with the leg frame **112** and carries at least a wheel **126**

rotatable relative to the wheel mount **124** about a wheel axis **126A** extending generally horizontally, whereby the wheel mount **124** is rotatable for changing a horizontal orientation of the wheel axis **126A**. Likewise, each wheel assembly **110** can include a wheel mount **128** that is pivotally connected with the leg frame **114** and carries at least a wheel **130** rotatable relative to the wheel mount **128** about a wheel axis **130A** extending generally horizontally, whereby the wheel mount **128** is rotatable for changing a horizontal orientation of the wheel axis **130A**. The wheel mounts **124** are spaced apart from the wheel mounts **128** along a lengthwise axis of the wheeled carrying apparatus **100** extending from a rear to a front thereof.

(18) In conjunction with FIG. 1, FIG. 2 is a cross-sectional view illustrating one wheel assembly **108**. Referring to FIGS. 1 and 2, the two wheel assemblies **108** can have a same construction. In each wheel assembly **108**, the wheel mount **124** can be pivotally connected with the leg frame **112** about a pivot axle **132** so that the wheel mount **124** and the wheel **126** carried therewith can rotate in unison about the pivot axle **132**. According to an example of construction, the wheel mount **124** can include a housing **134** that is pivotally connected with the leg frame **112** via the pivot axle **132**. A wheel axle **136** defining the wheel axis **126A** can be disposed through the housing **134** for pivotally connecting the wheel **126** with the wheel mount **124**. According to an example of construction, the wheel axle **136** can be disposed through a sliding part **138** assembled inside the housing **134** for vertical sliding movement, and a spring **140** can be respectively connected with the sliding part **138** and the housing **134**. The spring **140** can cushion a vertical displacement of the wheel **126** relative to the wheel mount **124** for dissipating undesirable shock energy during use.

(19) Referring to FIG. 2, a locking assembly **142** is provided to rotationally lock and unlock the wheel mount **124** with respect to the standing frame **104**. The locking assembly **142** can include a wheel mount latch **144** movably connected with the standing frame **104**, and a locking spring **146** connected with the wheel mount latch **144**. According to an example of construction, the wheel mount latch **144** can be slidably connected with the side segment **112A** of the leg frame **112**. More specifically, the side segment **112A** can have a cavity, and the wheel mount latch **144** can have an elongate shape slidably received in the cavity of the side segment **112A**, whereby the wheel mount latch **144** can slide to protrude outside the cavity or retract inward. The locking spring **146** can be disposed inside the cavity of the side segment **112A**, and can have two ends respectively connected with the side segment **112A** and the wheel mount latch **144**. The locking spring **146** can bias the wheel mount latch **144** to protrude outward for engagement with the wheel mount **124**.

(20) In the locking assembly **142**, the wheel mount latch **144** can slide downward to engage with the wheel mount **124** and thereby rotationally lock the wheel mount **124** with respect to the standing frame **104**, and slide upward to disengage from the wheel mount **124** to unlock the wheel mount **124** so that the wheel mount **124** can rotate relative to the standing frame **104**. The locking spring **146** can apply a biasing force for urging the wheel mount latch **144** to engage with the wheel mount **124**. The same locking assembly **142** described above may be provided for each of the two wheel assemblies **108**.

(21) In conjunction with FIG. 1, FIG. 3 is an enlarged perspective view illustrating one wheel assembly **110**. Referring to FIGS. 1 and 3, the wheel assembly **110** can have the same construction described previously, which can include the wheel mount **128** pivotally connected with the leg frame **114** so that the wheel mount **128** and the wheel **130** carried therewith can rotate in unison relative to the leg frame **114**. Moreover, a locking assembly **148** can be provided to rotationally lock and unlock the wheel mount **128** with respect to the standing frame **104**. The locking assembly **148** may have a construction similar to the locking assembly **142** of the wheel assembly **108**. According to an example of construction, the locking assembly **148** can include a wheel mount latch **150** movably connected with the standing frame **104**, and a locking spring **152** connected with the wheel mount latch **150**. For example, the wheel mount latch **150** may be slidably connected with the side segment **114A** of the leg frame **114**, and the locking spring **152** can have two ends respectively connected with the side segment **114A** and the wheel mount latch **150**. The

wheel mount latch **150** can thereby slide downward to engage with the wheel mount **128** and thereby rotationally lock the wheel mount **128** with respect to the standing frame **104**, and can slide upward to disengage and unlock the wheel mount **128** so that the wheel mount **128** can freely rotate relative to the standing frame **104** for changing the horizontal orientation of the wheel axis **130A**. The locking spring **152** can bias the wheel mount latch **150** to engage with the wheel mount **128**. (22) In conjunction with FIGS. 1-3, FIGS. 4-11 are schematic views illustrating further construction details that can be provided in the wheeled carrying apparatus **100**. Referring to FIGS. 1-9, the wheeled carrying apparatus **100** can further include an actuating assembly **160** and a release mechanism **162**.

(23) Referring to FIGS. 1-5, the actuating assembly **160** is respectively coupled to the handle frame **106** and the wheel mount latches **144** and **150**, and is operable independent of the release mechanism **162** to switch the wheel mount latches **144** and **150** between a locking and an unlocking state in accordance with the position of the handle frame **106**. According to an example of construction, the actuating assembly **160** can include a link actuator **164** coupled to the standing frame **104**, a linking part **166** coupling the wheel mount latch **144** to the link actuator **164**, and a linking part **168** coupling the wheel mount latch **150** to the link actuator **164**, the wheel mount latches **144** and **150** being respectively associated with the wheel assemblies **108** and **110** located at a same side (i.e., a left or right side) of the wheeled carrying apparatus **100**. The linking parts **166** and **168** can include one or more cable and are illustrated with phantom lines FIG. 4.

(24) The link actuator **164** can move in a first direction to pull the linking part **168** and urge the wheel mount latch **150** to disengage from the wheel mount **128** and concurrently relax the linking part **166** so that the wheel mount latch **144** can engage with the wheel mount **124** under the biasing force of the locking spring **146**, and in a second direction opposite to the first direction to pull the linking part **166** and urge the wheel mount latch **144** to disengage from the wheel mount **124** and concurrently relax the linking part **168** so that the wheel mount latch **150** can engage with the wheel mount **128** under the biasing force of the locking spring **152**. The link actuator **164** is movably linked to the handle frame **106** so that the link actuator **164** is movable in the first direction or the second direction in response to a rotation of the handle frame **106** relative to the standing frame **104** between the first position and the second position of different inclinations. In this embodiment, the link actuator **164** can be pivotally connected with the standing frame **104** about the pivot axis R, and can rotate as a unitary part relative to the standing frame **104** in the first and second direction. According to an example of construction, the side portion **106A** of the handle frame **106** can be fixedly connected with a shaft portion **170** extending along the pivot axis R, and the link actuator **164** can be disposed adjacent to the carrier bar **120** and can be rotationally coupled to the handle frame **106** by engaging the shaft portion **170** through a hole **164A** provided in the link actuator **164**, whereby the link actuator **164** and the handle frame **106** can rotate in unison about the pivot axis R relative to the standing frame **104**. Examples of suitable structures for the link actuator **164** may include, without limitation, a rotary plate, a ring and the like.

(25) Referring to FIGS. 1-5, the two linking parts **166** and **168** can include cable portions that are provided as separate cables or are defined from a single cable, and can be routed along the standing frame **104**. Suitable cables may include any elongate and flexible structures that can be tensioned for transmitting a drawing force, which may include, without limitation, metallic cables, cords, ropes, wires, strings, bands, strips, and the like. According to an example of construction, the two linking parts **166** and **168** can respectively include two cable portions that respectively couple the wheel mount latches **144** and **150** to the link actuator **164**. For example, the linking part **166** can include a cable portion that is anchored to the link actuator **164** via an attachment part **172** and has an end anchored to the wheel mount latch **144**, and the linking part **168** can include a cable portion that is anchored to the link actuator **164** via the attachment part **172** and has an end anchored to the wheel mount latch **150**. The linking part **166** can exemplarily extend from the wheel mount latch **144** along the side segment **112A** of the leg frame **112**, the side linking bar **116** and the carrier bar

120 to the link actuator **164**. The linking part **168** can exemplarily extend from the wheel mount latch **150** along the side segment **114A** of the leg frame **114**, the side linking bar **116** and the carrier bar **120** to the link actuator **164**. The linking parts **166** and **168** may extend substantially parallel to each other along the carrier bar **120**.

(26) According to an example of construction, the two linking parts **166** and **168** may be defined from a single cable that can thereby form an integral single linking part, which can be routed along the frame structure like previously described, and can have a first and a second end that are respectively anchored to the wheel mount latch **144** and the wheel mount latch **150** and an intermediate portion that loops around the link actuator **164** and is anchored to the link actuator **164** via the attachment part **172**.

(27) In the actuating assembly **160**, the link actuator **164** can thus rotate along with the handle frame **106** in a first direction to pull the linking part **168** (in particular the end thereof that is anchored to the wheel mount latch **150**) and thereby urge the wheel mount latch **150** to disengage and unlock the wheel mount **128** and concurrently relax the linking part **166** (in particular the end thereof that is anchored to the wheel mount latch **144**) so that the wheel mount latch **144** is allowed to engage and lock the wheel mount **124**. Moreover, the link actuator **164** can rotate along with the handle frame **106** in a second direction opposite to the first direction to pull the linking part **166** (in particular the end thereof that is anchored to the wheel mount latch **144**) and thereby urge the wheel mount latch **144** to disengage and unlock the wheel mount **124** and concurrently relax the linking part **168** (in particular the end thereof that is anchored to the wheel mount latch **150**) so that the wheel mount latch **150** is allowed to engage and lock the wheel mount **128**.

(28) Since the two wheels assemblies **108** and **110** are provided at each of the left and right side of the wheeled carrying apparatus **100**, the actuating assembly **160** can be symmetrically arranged at the left and right side.

(29) Referring to FIGS. **1-3** and **5-8**, the release mechanism **162** includes an operating part **174** carried with the handle frame **106**, and is configured so that the operating part **174** can be operatively coupled to one of the wheel mount latches **144** and **150** and decoupled from the other one of the wheel mount latches **144** and **150** depending on whether the handle frame **106** is in the first position inclined toward the side of the leg frame **112** or the second position inclined toward the side of the leg frame **114**. For example, the operating part **174** is operable to cause the wheel mount latch **144** to disengage from the wheel mount **124** and meanwhile is operatively decoupled from the wheel mount latch **150** when the handle frame **106** is in the first position. When the handle frame **106** is in the second position, the operating part **174** is operable to cause the wheel mount latch **150** to disengage from the wheel mount **128** and meanwhile is operatively decoupled from the wheel mount latch **144**. To this end, an embodiment of the release mechanism **162** can include the operating part **174**, two link actuators **176** and **178** operable independently of each other, a linking part **180** coupling the wheel mount latch **144** to the link actuator **176**, a linking part **182** coupling the wheel mount latch **150** to the link actuator **178**, a driving part **184** carried with the handle frame **106**, and a linking part **186** coupling the driving part **184** to the operating part **174**. The linking parts **180**, **182** and **186** may include cables and are illustrated with phantom lines in FIG. **6**. Suitable cables may include any elongate and flexible structures that can be tensioned for transmitting a drawing force, which may include, without limitation, metallic cables, cords, ropes, wires, strings, bands, strips, and the like.

(30) Referring to FIGS. **1-3** and **5-7**, the link actuator **176** is movably connected with the standing frame **104**, and can respectively move in one direction to pull the linking part **180** and thereby urge the wheel mount latch **144** to disengage from the wheel mount **124** and in an opposite direction to relax the linking part **180** so that the wheel mount latch **144** can engage with the wheel mount **124**. The link actuator **176** can be placed at a position that is adjacent to the side portion **106A** of the handle frame **106** when the handle frame **106** is in the first position. According to an example of construction, the link actuator **176** can be movably connected and carried with the carrier bar **120**

of the standing frame **104**, and can have a protrusion **176A** projecting sideways. For example, the link actuator **176** may be slidably connected with the carrier bar **120**, and can slide upward to pull the linking part **180** and thereby urge the wheel mount latch **144** to disengage from the wheel mount **124** and slide downward to relax the linking part **180** so that the wheel mount latch **144** can engage with the wheel mount **124**.

(31) The linking part **180** can include a cable, which may have one end anchored to the link actuator **176** and another end anchored to the wheel mount latch **144**. The linking part **180** may extend substantially parallel to the two linking parts **166** and **168** of the actuating assembly **160** along a portion of the carrier bar **120**.

(32) Referring to FIGS. 5-7, the linking part **180** may extend along a portion of the side segment **112A** of the leg frame **112**, a portion of the side linking bar **116**, a portion of the carrier bar **120**, and can loop around a pulley **188** that is pivotally connected with the carrier bar **120**. The pulley **188** may be disposed in a region of the carrier bar **120** that is located between the link actuator **164** of the actuating assembly **160** and the link actuator **176**.

(33) FIGS. **10** and **11** illustrate a variant construction in which the linking part **180** can operate like previously described but is routed along a different path. Referring to FIGS. **10** and **11**, the linking part **180** may extend along a portion of the side segment **112A** of the leg frame **112**, the bar **122A** and a portion of the carrier bar **120** for connecting with the link actuator **176**. In this variant construction, the pulley **188** can be omitted, and the link actuator **176** can be connected with a coupling rod **196** provided on the carrier bar **120**. The coupling rod **196** can slide along with the link actuator **176** to pull the linking part **180** and urge the wheel mount latch **144** to disengage from the wheel mount **124**.

(34) Referring to FIGS. **5** and **7**, the release mechanism **162** can further include a spring **190** connected with the link actuator **176**. For example, the spring **190** may have one end connected with the link actuator **176**, and another end connected with an anchor structure fixedly connected with the carrier bar **120**. The spring **190** can bias the link actuator **176** in a direction that urges the link actuator **176** in contact with the driving part **184**.

(35) In the variant construction illustrated in FIG. **11**, the spring **190** can operate in a same manner, and may be connected between the link actuator **176** and an anchor structure provided on the coupling rod **196**.

(36) Referring to FIGS. **6** and **8**, the link actuator **178** is movably connected with the standing frame **104**, and can respectively move in one direction to pull the linking part **182** and thereby urge the wheel mount latch **150** to disengage from the wheel mount **128** and in an opposite direction to relax the linking part **182** so that the wheel mount latch **150** can engage with the wheel mount **128**. The link actuator **178** can be placed at a position that is adjacent to the side portion **106A** of the handle frame **106** when the handle frame **106** is in the second position. According to an example of construction, the link actuator **178** can be movably connected and carried with the side segment **114A** of the leg frame **114**, and can have a protrusion **178A** projecting sideways. For example, the link actuator **178** may be pivotally connected with a bracket **191** that is fixedly attached to the side segment **114A** of the leg frame **114**, and can rotate in one direction to pull the linking part **182** and thereby urge the wheel mount latch **150** to disengage from the wheel mount **128** and rotate in an opposite direction to relax the linking part **182** so that the wheel mount latch **150** can engage with the wheel mount **128**. It will be appreciated that the aforementioned pivotal connection is only an example of movably connecting the link actuator **178** with the side segment **114A** of the leg frame **114**, and other movable connections of the link actuator **178** may be suitable for pulling or relaxing the linking part **182**. For example, a variant construction may have the link actuator **178** slidably connected with the side segment **114A** of the leg frame **114** so that the link actuator **178** can slide for pulling or relaxing the linking part **182**. According to an example of construction, the linking part **182** can include a cable, which may have one end anchored to the link actuator **178** and another end anchored to the wheel mount latch **150**. According to an example of construction, the

linking part **182** may extend along a portion of the side segment **114A** of the leg frame **114** from the wheel mount latch **150** to the link actuator **178**.

(37) Referring to FIGS. 5-8, the driving part **184** is movably connected and carried with the handle frame **106**, and is coupled to the operating part **174** via the linking part **186**. According to an example of construction, the driving part **184** can be slidably connected with the side portion **106A** of the handle frame **106** in a region below the side linking bar **116**, and can have a protrusion **184A** projecting sideways. Accordingly, the driving part **184** is movable along with the handle frame **106** when the handle frame **106** rotates relative to the standing frame **104** between the first and second position of different inclinations. When the handle frame **106** is in the first position inclined toward the side of the leg frame **112**, the driving part **184** can be located adjacent to the link actuator **176** and displaced away from the link actuator **178**. When the handle frame **106** is in the second position inclined toward the side of the leg frame **114**, the driving part **184** can be located adjacent to the link actuator **178** and displaced away from the link actuator **176**.

(38) According to an example of construction, the linking part **186** can include a cable, which may have one end anchored to the driving part **184** and another end anchored to the operating part **174**. For example, the operating part **174** may be provided on the grip portion **106B** of the handle frame **106**, and the linking part **186** can extend along the side portion **106A** of the handle frame **106** and can have two opposite ends respectively anchored to the operating part **174** and the driving part **184**. Accordingly, the operating part **174** can be operable to cause the driving part **184** to move relative to the side portion **106A** of the handle frame **106** and selectively urge either of the link actuator **176** and **178** in movement for unlocking either of the wheel mount latch **144** and **150**. For example, the operating part **174** is operable to cause the wheel mount latch **144** to disengage from the wheel mount **124** via a contact between the driving part **184** and the link actuator **176** when the handle frame **106** is in the first position, and the operating part **174** is operable to cause the wheel mount latch **150** to disengage from the wheel mount **128** via a contact between the driving part **184** and the link actuator **178** when the handle frame **106** is in the second position.

(39) FIG. 9 is an enlarged view illustrating construction details of the operating part **174**. Referring to FIG. 9, the operating part **174** can include a barrel **192** and a button **194**. The barrel **192** can be pivotally connected with the grip portion **106B** of the handle frame **106**, and the linking part **186** can have an end anchored to the barrel **192**. The button **194** can be slidably connected with the grip portion **106B** and pivotally connected with the barrel **192**. With this construction, the button **194** can be pressed to urge the barrel **192** to rotate for pulling the linking part **186** so that the driving part **184** is urged in movement relative to the handle frame **106**.

(40) Since the two wheels assemblies **108** and **110** are provided at each of the left and right side of the wheeled carrying apparatus **100**, the release mechanism **162** can be symmetrically arranged at the left and right side and likewise coupled to the operating part **174**.

(41) In the wheeled carrying apparatus **100**, the handle frame **106** can be inclined toward the side of the leg frame **112** in the first position and toward the side of the leg frame **114** in the second position. The link actuator **164** of the actuating assembly **160** can rotate along with the handle frame **106** in the first direction for pulling the linking part **168** and relaxing the linking part **166** when the handle frame **106** rotates from the second position to the first position, which can cause the wheel mount latch **150** to disengage and unlock the wheel mount **128** and cause the wheel mount latch **144** to engage and lock the wheel mount **124**. Conversely, the link actuator **164** can rotate along with the handle frame **106** in the second direction for pulling the linking part **166** and relaxing the linking part **168** when the handle frame **106** rotates from the first position to the second position, which can cause the wheel mount latch **144** to disengage and unlock the wheel mount **124** and cause the wheel mount latch **150** to engage and lock the wheel mount **128**.

(42) While the handle frame **106** is in the first position and the operating part **174** is released, the wheel mount latch **144** is engaged with the wheel mount **124** to lock the wheel mount **124** to the standing frame **104**, and the wheel mount latch **150** is kept disengaged from the wheel mount **128**

by a distance so that the wheel mount **128** is unlocked for free rotation relative to the standing frame **104**. For unlocking the wheel mounts **124** and **128** while the handle frame **106** is in the first position, a caregiver can actuate the operating part **174** to pull the linking part **186**, which causes the driving part **184** to slide upward and urge the link actuator **176** in movement against the biasing force of the spring **190** via a contact between the protrusion **184A** of the driving part **184** and the protrusion **176A** of the link actuator **176**. As a result, the link actuator **176** can pull the linking part **180** and thereby urge the wheel mount latch **144** to disengage from the wheel mount **124**.

Meanwhile, the wheel mount latch **150** is not affected by the actuation of the operating part **174** and can remain in position disengaged from the wheel mount **128** by the same distance.

Accordingly, all of the wheel mounts **124** and **128** can be unlocked at the same time in the first position of the handle frame **106**.

(43) While the handle frame **106** is in the second position and the operating part **174** is released, the wheel mount latch **150** is engaged with the wheel mount **128** to lock the wheel mount **128** to the standing frame **104**, and the wheel mount latch **144** is kept disengaged from the wheel mount **124** by a distance so that the wheel mount **124** is unlocked for free rotation relative to the standing frame **104**. For unlocking the wheel mounts **124** and **128** while the handle frame **106** is in the second position, a caregiver can actuate the operating part **174** to pull the linking part **186**, which causes the driving part **184** to slide upward and urge the link actuator **178** in movement via a contact between the protrusion **184A** of the driving part **184** and the protrusion **178A** of the link actuator **176**. As a result, the link actuator **178** can pull the linking part **182** and thereby urge the wheel mount latch **150** to disengage from the wheel mount **128**. Meanwhile, the wheel mount latch **144** is not affected by the actuation of the operating part **174** and can remain in position disengaged from the wheel mount **124** by the same distance. Accordingly, all of the wheel mounts **124** and **128** can be unlocked at the same time in the second position of the handle frame **106**.

(44) Since the operating part **174** is operatively coupled to only one of the wheel mount latches **144** and **150** at a time at each of the left and right side, the actuation for having all of the wheel mounts **124** and **128** unlocked at the same time would require a driving force that is relatively smaller.

Therefore, less effort would be required to actuate the operating part **174**, and the component parts of the release mechanism **162** can be subjected to less mechanical stress.

(45) Advantages of the wheeled carrying apparatuses described herein include the ability to lock and unlock wheel assemblies with respect to a standing frame in a flexible manner. According to the needs, some of the wheel assemblies are locked while others are unlocked, or all of the wheel assemblies are unlocked at the same time as desired. Accordingly, the wheeled carrying apparatuses can be more flexible in use and offer better maneuverability.

(46) Realization of the wheeled carrying apparatuses has been described in the context of particular embodiments. These embodiments are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. These and other variations, modifications, additions, and improvements may fall within the scope of the inventions as defined in the claims that follow.

Claims

1. A wheeled carrying apparatus comprising: a standing frame and a handle frame pivotally coupled to each other, the handle frame being rotatable relative to the standing frame; a first wheel mount carrying a first wheel and pivotally connected with the standing frame, the first wheel being rotatable relative to the first wheel mount about a first wheel axis; a locking assembly connected with the standing frame, the locking assembly being configured to rotationally lock the first wheel mount with respect to the standing frame; and a release mechanism synchronously moved with the handle to couple or decouple with the locking assembly, wherein the release mechanism includes a first portion provided on the standing frame and coupled to the locking assembly, and a second

portion carried with the handle frame, the second portion being movable along with the handle frame during rotation of the handle frame between a first position and a second position; wherein the second portion is operable to actuate the first portion and cause the locking assembly to unlock the first wheel mount when the handle frame is in the first position, and the second portion is decoupled from the first portion so that the second portion is inoperable to cause the locking assembly to unlock the first wheel mount when the handle frame is in the second position.

2. The wheeled carrying apparatus according to claim 1, wherein causing the locking assembly to unlock the first wheel mount allows the first wheel mount to rotate relative to the standing frame for changing an orientation of the first wheel axis.

3. The wheeled carrying apparatus according to claim 1, wherein the second portion is located adjacent to the first portion so that the second portion is operable to actuate the first portion and cause the locking assembly to unlock the first wheel mount when the handle frame is in the first position, and the second portion is displaced away and disconnected from the first portion so that the second portion is inoperable to cause the locking assembly to unlock the first wheel mount when the handle frame is in the second position.

4. The wheeled carrying apparatus according to claim 1, wherein the locking assembly includes a first wheel mount latch movably connected with the standing frame, the first wheel mount latch being movable to engage with the first wheel mount for rotationally locking the first wheel mount with respect to the standing frame and to disengage from the first wheel mount for rotation of the first wheel mount relative to the standing frame.

5. The wheeled carrying apparatus according to claim 1, wherein the release mechanism is symmetrically arranged on the handle frame.

6. The wheeled carrying apparatus according to claim 1, wherein the release mechanism is coupled to a distal end of a side portion of the handle frame.

7. The wheeled carrying apparatus according to claim 6, wherein the release mechanism extends in a direction parallel to an extension direction of the side portion of the handle frame.

8. A wheeled carrying apparatus comprising: a standing frame and a handle frame pivotally coupled to each other, the handle frame being rotatable relative to the standing frame; a first wheel mount carrying a first wheel and pivotally connected with the standing frame, the first wheel being rotatable relative to the first wheel mount about a first wheel axis; a locking assembly connected with the standing frame, the locking assembly being configured to rotationally lock the first wheel mount with respect to the standing frame; and a release mechanism synchronously moved with the handle to couple or decouple with the locking assembly, wherein the release mechanism includes a first portion provided on the standing frame and coupled to the locking assembly, and a second portion carried with the handle frame, the second portion being movable along with the handle frame during rotation of the handle frame between a first position and a second position; the first portion includes a link actuator that is coupled to the locking assembly via a first linking part, and the second portion includes a driving part and an operating part that are coupled to each other via a second linking part, the driving part being located adjacent to the link actuator when the handle frame is in the first position and is displaced away from the link actuator when the handle frame is in the second position.

9. The wheeled carrying apparatus according to claim 8, wherein the driving part is movably connected with a side portion of the handle frame.

10. The wheeled carrying apparatus according to claim 8, wherein the handle frame has a side portion and a grip portion connected with each other, the driving part being movably connected with the side portion, and the operating part being provided on the grip portion.

11. The wheeled carrying apparatus according to claim 8, wherein any one of the first linking part and the second linking part includes a cable.

12. The wheeled carrying apparatus according to claim 8, wherein the first linking part loops around a pulley.

13. The wheeled carrying apparatus according to claim 8, wherein the link actuator is connected with a spring, the spring biasing the link actuator in a direction that urges the link actuator in contact with the driving part.

14. The wheeled carrying apparatus according to claim 8, wherein the standing frame includes a carrier bar pivotally connected with the handle frame, the link actuator being movably connected and carried with the carrier bar.
